

Electoral Accountability and Economic Policy

Based on Besley & Case (1995)

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- A central question in political economy: **do electoral incentives discipline politicians?**
- Do politicians act in the public interest, or do they use political power for personal gain?
- The desire to maintain a reputation is often thought to keep politicians in check
- Officials who care to run again must act in voters' interest to merit reelection
- But what happens when reelection is no longer possible?
- **Besley & Case (1995)** exploit gubernatorial term limits in U.S. states to test this

Research Question

Central Question

Does the removal of reelection incentives — through a binding term limit — alter economic policy choices?

- If incumbents care about reputation, they should behave differently when they cannot run again
- A binding term limit removes the future electoral reward that keeps politicians in check, allowing us to observe what happens when accountability disappears.
- This allows a clean test of the **political agency model**

Theoretical Framework: Reputation-Building Model

- Based on Banks & Sundaram (1993)
- Each governor has an unobservable **type** $\omega \in \{\omega_1, \dots, \omega_N\}$
- Takes unobservable **effort** α , which affects policy outcome r
- Voters observe r and use it to infer the governor's type via Bayes' rule

Two Regimes

- 1 **Term-limited (lame duck)**: maximizes immediate payoffs — puts in less effort
- 2 **Reelection-eligible**: puts in extra effort to build reputation and secure reelection

Short-run (term-limited) decision:

$$\alpha_s(\omega) = \arg \max_{\alpha} \{u(\alpha, \omega) \mid \alpha \in [\underline{\alpha}, \bar{\alpha}]\}$$

Long-run (reelection-eligible) decision:

$$\alpha_l(\omega) = \arg \max_{\alpha} \{u(\alpha, \omega) + \delta \Pr[r \in R(\sigma) \mid \alpha] u(\alpha_s(\omega), \omega)\}$$

Key Proposition

Incumbents eligible for reelection put in **more effort** than term-limited ones. Those in their last term give lower payoffs to voters, on average.

Estimated equation:

$$P_{st} = \zeta_s + \psi_t + \gamma T_{st} + \alpha Z_{st} + \varepsilon_{st}$$

where:

- P_{st} : policy outcome (taxes, spending, minimum wage) in state s , year t
- ζ_s : state fixed effects
- ψ_t : year fixed effects
- $T_{st} = 1$ if governor cannot run again (binding term limit)
- Z_{st} : controls (income per capita, demographics, population)
- γ : **coefficient of interest**

If $\gamma = 0$, the reputation-building model is not supported.

Main Results

Outcome	Coefficient on T_{st}	Significance
Sales taxes per capita	+7.86	**
Income taxes per capita	+8.74	**
Corporate taxes per capita	+0.57	—
Total taxes per capita	+6.71	—
State expenditure per capita	+14.38	**
State minimum wage	−0.14	**

** $p < 0.05$. All variables per capita in 1982 dollars.

- Term-limited governors raise taxes and spending
- They reduce minimum wages — consistent with less resistance to special interests

Key Finding

The term-limit effect is **driven entirely by Democratic governors**.

- Democratic lame ducks: raise sales, income, and corporate taxes significantly
- Republican lame ducks: no significant effect on taxes or spending
- Suggests **incomplete party discipline** — individual incentives matter
- Fiscal cycle interpretation: Democrats hold spending down in first term to get reelected, then increase it in the lame-duck term

The Fiscal Cycle

- The main effect of term limits is not a permanent shift but a **fiscal cycle**
- Governors hold spending **below** the state mean in their first term
- Governors hold spending **above** the state mean in their lame-duck term
- No significant within-term variation in behavior — the shift is sharp at the term boundary
- Effect on **state income per capita**: lame-duck Democratic governors are associated with a 1–2% reduction in state income — the efficiency cost of the fiscal cycle

Relevance for Our Paper

Besley & Case (1995) establish:

- ① Reelection incentives discipline politicians' fiscal behavior
- ② Term limits provide exogenous variation in electoral incentives
- ③ The effect operates through reputation-building, not party discipline

Our contribution

We apply this framework to **Brazilian municipalities** in São Paulo, asking whether reelection eligibility affects **spending composition** — not just the level of taxes and spending — and whether this effect is moderated by **fiscal discretion**.

Brazilian Municipal Elections

- Brazilian municipalities hold **direct elections** for mayor every four years
- Elections are held simultaneously across all municipalities — in our sample, the relevant elections occurred in **2016** and **2020**
- Voter turnout is high by international standards, partly due to **compulsory voting** for citizens aged 18–70
- São Paulo state alone has **645 municipalities**, ranging from the capital with over 12 million inhabitants to small rural towns with fewer than 1,000 residents

Term Limits in Brazilian Municipalities

- Brazilian law limits mayors to **two consecutive terms** in office
- A mayor elected in 2016 for the **first time** is eligible to run again in 2020
- A mayor elected in 2016 who was **already serving their first term** since 2012 faces a binding term limit and cannot run in 2020
- This generates clean variation in reelection incentives **within the same election cycle** across municipalities

Identification

Comparing mayors who can run again in 2020 against those who cannot, conditional on controls, isolates the causal effect of reelection incentives on spending behavior.

Who Controls Municipal Spending?

- Brazilian municipalities operate under a **separation of powers**: the *Câmara dos Vereadores* (city council) approves laws and the annual budget, while the **mayor executes it**
- In practice, the mayor has **substantial discretion** over spending composition through several channels:
 - ① **Budget proposal**: the mayor drafts the annual budget (*Lei Orçamentária Anual*), which the council typically approves with minor modifications
 - ② **Execution discretion**: the mayor controls the *timing* and *allocation* of expenditures within approved categories
 - ③ **Supplementary credits**: the mayor can open *créditos suplementares* to reallocate spending across categories without full council approval
 - ④ **Administrative decisions**: hiring, procurement, and contracting decisions — which shape spending composition in practice — are executive prerogatives

The Limits of Mayoral Discretion

- Not all spending is equally subject to mayoral discretion
- A large share of municipal revenue is **constitutionally earmarked**:
 - **FUNDEB**: at least 25% of tax revenue must be spent on education
 - **SUS transfers**: federal health transfers are earmarked exclusively to health spending
 - **Convênios**: federal and state grants are tied to specific programs
- What remains after earmarked transfers is **treasury revenue** (*tesouro*) — the discretionary budget the mayor truly controls
- This motivates our key moderating variable: the **share of total spending funded by discretionary treasury resources**, D_{it} , which measures how much fiscal power the mayor actually has

Our Research Question

Central Question

Do reelection incentives affect how Brazilian mayors allocate their **discretionary** budget across spending categories?

- We follow Besley & Case (1995) in using **term limits** as a source of exogenous variation in reelection incentives
- Our contribution is to focus on **spending composition** rather than aggregate levels, and to explicitly account for **fiscal discretion** as a moderating force
- The Brazilian context is particularly informative because **heavy earmarking** limits mayoral discretion — if political budget cycles exist, they should be concentrated in the discretionary budget

- **Municipal expenditure data:** Tribunal de Contas do Estado de São Paulo (TCESP)
 - Annual expenditure records by spending category and funding source
 - 643 municipalities, 2017–2024
- **Electoral data:** Tribunal Superior Eleitoral (TSE)
 - Candidate-level data for 2016 and 2020 municipal elections
 - Reelection eligibility, vote shares, number of candidates
- **Socioeconomic controls:** IBGE Census 2010 and Atlas Brasil
 - Population, PIB per capita, age distribution
 - Municipal HDI, sanitation access, infant mortality
- All socioeconomic controls are measured in 2010, eliminating reverse causality concerns.

Dependent Variable: Discretionary Spending Share

- Standard measures of municipal spending conflate **mayoral choices** with **mechanically earmarked transfers**
- We focus instead on the **discretionary spending share**:

Dependent Variable

$$s_{ikt}^d = \frac{\text{Discretionary spending in category } k_{it}}{\text{Total discretionary spending}_{it}} \times 100$$

- **Discretionary spending** = expenditure funded by *tesouro* or *recursos próprios da administração indireta*
- **Earmarked spending** = federal/state transfers and special funds (*FUNDEB, SUS, convênios*)
- s_{ikt}^d measures **how much of total discretionary money did the mayor choose to put in a given category** — a genuine revealed preference for spending priority.

Key Variables

- **Treatment variable** $R_{it} \in \{0, 1\}$:
- $R_{it} \in \{0, 1\}$ varies at the **municipality-term** level:
 - For 2017–2020: $R_{i_1} = 1$ if the 2016 winner was in their **first term**, eligible to run in 2020; $R_{i_1} = 0$ if they had already served since 2012
 - For 2021–2024: $R_{i_2} = 1$ if the 2020 winner was in their **first term**, eligible to run in 2024; $R_{i_2} = 0$ if they had already served since 2016
- Predetermined relative to spending: determined by prior election outcomes before each term begins
- **Moderating variable** D_{it} :
 - Discretionary share of total expenditure:

$$D_{it} = \frac{\text{Total discretionary spending}_{it}}{\text{Total spending}_{it}} \times 100$$

- Measures **mayoral fiscal power** — how much of the total budget the mayor truly controls
- Varies at the municipality-year level as federal transfer rules and revenues change

Regression Equation

Estimated Model

$$s_{ikt}^d = \beta_0 + \beta_1 R_{it} + \beta_2 \tilde{D}_{it} + \beta_3 (R_{it} \times \tilde{D}_{it}) + \beta_4 E_{it} + \beta_5 N_{it} + \mathbf{X}_i' \gamma + \mu_t + \varepsilon_{it}$$

- s_{ikt}^d : discretionary spending share in category k , municipality i , year t
- R_{it} : reelection eligibility — varies at municipality-term level
- $\tilde{D}_{it}$: demeaned discretionary share of total expenditure
- $R_{it} \times \tilde{D}_{it}$: interaction — does greater fiscal discretion amplify the effect of reelection incentives on spending composition?
- E_{it} : total expenditure per capita
- N_{it} : number of candidates in the relevant election — varies at municipality-term level
- $\mathbf{X}_i$: 2010 Census controls (PIB per capita, IDHM, share of children)
- μ_t : year fixed effects

Coefficients of Interest

- β_1 — **main effect of reelection eligibility:**
 - Effect of reelection eligibility on discretionary spending composition **at the sample mean of fiscal discretion** ($\bar{D} = 71.4\%$)
 - If $\beta_1 \neq 0$: reelection incentives distort discretionary spending composition
 - If $\beta_1 = 0$: no political budget cycle in discretionary spending
- β_2 — **effect of fiscal discretion on spending composition:**
 - How spending shares change as fiscal discretion moves **above or below the sample mean** of 71.4%
 - Positive β_2 : above average fiscal discretion $\rightarrow$ more spending in that category
 - Negative β_2 : above average fiscal discretion $\rightarrow$ less spending in that category

- β_3 — **moderation effect of fiscal discretion on reelection incentives:**
 - How the spending gap between reelection-eligible and term-limited mayors changes as fiscal discretion moves above or below its sample mean
 - Negative β_3 : gap **narrows** for above-average discretion municipalities
 - Positive β_3 : gap **widens** for above-average discretion municipalities

Spending Categories

- We estimate the model separately for **six functional spending categories** from TCESP data

Category	Description
Health (<i>Saúde</i>)	Primary healthcare, hospitals, and public health programs
Education (<i>Educação</i>)	Schools, teachers, and educational infrastructure
Urbanismo	Urban infrastructure, roads, and public works
Social Assistance	Welfare programs and social protection
Administration	General government overhead and bureaucracy
Transport	Public transportation and road maintenance

- Health and education are heavily constrained by **constitutional earmarking** (SUS, FUNDEB) — the reelection effect and its moderation by fiscal discretion are expected to be concentrated in the remaining categories where mayors have genuine room to maneuver

- We estimate the model separately for six spending categories
- Results are organized around three findings:
 - ① **Reelection eligibility** (β_1) significantly affects discretionary spending in **urbanismo** and **administration**
 - ② **Fiscal discretion** (β_2) shapes spending composition independently of electoral incentives in **social assistance**, **transport**, and **urbanismo**
 - ③ The **moderation effect** (β_3) is statistically significant in **social assistance** and **administration**, but economically negligible
- Health, education, and transport show **no significant reelection effect** — consistent with constitutional earmarking limiting mayoral discretion in these categories

Finding 1: Reelection Eligibility and Spending Composition

Category	Coefficient on R_{it} (β_1)			
	Estimate	Std. Error	T-stat	P-value
Health	-0.093	0.279	-0.34	0.738
Education	0.065	0.199	0.32	0.746
Urbanismo	-0.963***	0.299	-3.22	0.001
Social Assist.	0.080	0.100	0.80	0.425
Administration	1.027***	0.373	2.75	0.006
Transport	0.023	0.201	0.11	0.910

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Clustered SE at municipality level.

- Reelection eligibility affects spending composition **only through urbanismo and administration**

Finding 1: Reelection Eligibility and Urbanismo

Result

Reelection-eligible mayors allocate **0.96 percentage points less** of their discretionary budget to urbanismo relative to term-limited mayors ($\beta_1 = -0.963$, $p = 0.001$)

- Urbanismo covers **urban infrastructure, roads, and public works** — a category historically associated with corruption in Brazilian municipalities
- We interpret the **negative β_1** as evidence of **electoral accountability constraining opportunistic spending**: reelection-eligible mayors restrain infrastructure spending because corruption in public works carries electoral risk before the next election

Finding 1: Reelection Eligibility and Urbanismo

Result

Reelection-eligible mayors allocate **0.96 percentage points less** of their discretionary budget to urbanismo relative to term-limited mayors ($\beta_1 = -0.963$, $p = 0.001$)

- Term-limited mayors, facing **no future electoral punishment**, increase discretionary urbanismo spending — consistent with rent extraction through inflated construction contracts, a well-documented phenomenon in Brazilian public administration
- This is consistent with the **reputation-building model**: accountability disciplines mayors when reelection is at stake, and opportunistic behavior emerges when it is not

Finding 1: Reelection Eligibility and Administration

Result

Reelection-eligible mayors allocate **1.03 percentage points more** of their discretionary budget to administration relative to term-limited mayors ($\beta_1 = 1.027$, $p = 0.006$)

- Administration covers **general government overhead**, including public hiring and bureaucratic expenditure
- We interpret this as evidence of **political patronage**: reelection-seeking mayors strategically expand the municipal payroll to generate electoral loyalty

Finding 1: Reelection Eligibility and Administration

Result

Reelection-eligible mayors allocate **1.03 percentage points more** of their discretionary budget to administration relative to term-limited mayors ($\beta_1 = 1.027$, $p = 0.006$)

- This is consistent with the broader literature on **political clientelism** in Brazilian municipalities, where public employment is a well-documented tool of electoral mobilization
- Importantly, this effect is identified **within the discretionary budget** — it reflects a genuine mayoral choice to prioritize administrative hiring over service delivery when reelection is at stake

Finding 2: Fiscal Discretion and Spending Composition

Result

The discretionary share of total expenditure ($\tilde{D}_{it}$) significantly shapes spending composition in three categories, independently of electoral incentives

Category	Estimate	Std. Error	T-stat	P-value
Social Assist.	+0.054***	0.015	3.63	0.000
Transport	+0.069***	0.025	2.74	0.006
Urbanismo	−0.095**	0.041	−2.33	0.020

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Finding 2: Fiscal Discretion and Spending Composition

Result

The discretionary share of total expenditure ($\tilde{D}_{it}$) significantly shapes spending composition in three categories, independently of electoral incentives

- Municipalities with **above average fiscal discretion** allocate proportionally more to social assistance and transport — when mayors have more budget freedom than the typical municipality, these categories reflect **genuine mayoral spending preferences**
- Municipalities with **above average fiscal discretion** allocate proportionally **less** to urbanismo — when mayors have more fiscal freedom, infrastructure competes less successfully for discretionary funds as other categories absorb the additional room to maneuver

Finding 2: Fiscal Discretion and Spending Composition

Result

The discretionary share of total expenditure ($\tilde{D}_{it}$) significantly shapes spending composition in three categories, independently of electoral incentives

- **Robustness check:** Fiscal discretion is **not simply a proxy for municipal wealth** — the correlation between $\tilde{D}_{it}$ and PIB per capita is only 0.12, and with IDHM only 0.21

Finding 3: The Moderation Effect

Result

The interaction $R_{it} \times \tilde{D}_{it}$ reveals **heterogeneity in the form of electoral manipulation** depending on the level of fiscal discretion

Category	Estimate	Std. Error	T-stat	P-value
Health	+0.079*	0.047	1.68	0.094
Social Assist.	-0.040***	0.014	-2.84	0.005
Administration	-0.149**	0.057	-2.60	0.009

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Finding 3: Conditional Clientelism

Unified Interpretation

The form of electoral manipulation depends on **how much fiscal room a mayor has**

- **Mayors with below average fiscal discretion** rely on **two electoral instruments simultaneously**:
 - **Patronage hiring** — expanding the municipal payroll generates electoral loyalty when discretionary funds are scarce, with the effect growing stronger as discretion falls below the mean
 - **Social assistance** — targeting direct transfers to voters becomes the preferred electoral tool when fiscal room is tight
- **Mayors with above average fiscal discretion** show **attenuated effects** in both categories — at one standard deviation above the mean, the administration effect (+1.027 pp) is nearly fully offset (−1.12 pp), suggesting that patronage hiring becomes unnecessary when mayors have ample budget room

Finding 3: Conditional Clientelism

Unified Interpretation

The form of electoral manipulation depends on **how much fiscal room a mayor has**

- This is consistent with a **conditional clientelism** story: fiscal constraint shapes the instruments of electoral manipulation, with cash-strapped mayors relying more heavily on patronage and direct transfers

The Null Results: Health and Education

- Reelection eligibility has **no significant effect** on discretionary spending shares for health or education
- This is consistent with **constitutional earmarking** constraining mayoral choices even within the discretionary budget:
 - FUNDEB mandates minimum education spending — mayors face political and legal costs from deviating
 - SUS transfers create institutional pressures to maintain health spending levels
- The null results are a **substantive finding**, not a failure of the model — they confirm that institutional constraints on spending are binding even at the level of discretionary allocation
- This contributes to the literature on **fiscal federalism and political accountability**: earmarking effectively insulates key social spending from electoral manipulation, even when mayors have discretionary resources available